



Ansible Variables and Host Management - 2

Ansible inventory defines hosts and groups of hosts for task execution. Hosts can be written in ranges, e.g. `db[01:05].example.com` or alphabetic `db[a:f].example.com`. Groups can be nested with `:children` to allow hierarchical host management.

Grouping and Subgroups

Regional Groups

Instead of using regex to select hosts, you can structure inventory with groups like:

```
[web_tokyo]
web1.example.com
[db_tokyo]
db1.example.com
```

Hierarchical Structure

Use :children to create nested groups:

```
[webservers:children]
web_tokyo
[tokyo:children]
web_tokyo
db_tokyo
```

Inventory Directories

Instead of a single inventory file, you can store multiple files in a directory (inventory/) and Ansible merges them. File load order is ASCII-based, so prefix like 01-, 02- helps control order.

Example structure:

```
inventory/  
├── 01-webservers  
├── 02-databases  
└── 03-others
```



Dynamic Inventories



Static Inventories

Static inventories are fine for stable environments.



Dynamic Inventories

Dynamic inventories are better for cloud or containerised environments where hosts change frequently.



Inventory Plug-ins

Ansible provides many inventory plug-ins (ansible-doc -t inventory -l).

Dynamic Inventory with Docker and Podman

Dynamic Inventory with Docker

- Install Docker collection `ansible-galaxy collection install community.docker`
- Run container `docker run -it --rm --name test redhat/ubi8`
- Create inventory config (main_docker.yaml) plugin:
`community.docker.docker_containers docker_host: unix:///var/run/docker.sock connection_type: docker-cli`
- Test inventory `ansible-inventory -i main_docker.yaml --graph`

Dynamic Inventory with Podman

- Install Podman collection `ansible-galaxy collection install containers.podman`
- Run container `podman run -it --rm --name test docker.io/library/ubuntu:latest`
- Create inventory config (main_podman.yaml) plugin:
`containers.podman.podman_containers podman_host: unix:///run/podman/podman.sock connection_type: podman-cli`
- Test inventory `ansible-inventory -i main_podman.yaml --graph`

Conclusion



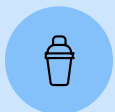
Inventory Types

Inventories can be static (INI/YAML) or dynamic (via plug-ins).



Docker Integration

Docker: `community.docker.docker_containers` plug-in.



Podman Integration

Podman: `containers.podman.podman_containers` plug-in.



Automatic Discovery

Both allow automatically discovering running containers as Ansible hosts.